[INCH-POUND]
A-A-59440/7A
23 May 2005
SUPERSEDING
A-A-59440/7
13 August 1999

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

COCK, SHUT-OFF, SCREW STEM, THREADED TYPE, 150 P.S.I., BRASS, WATER, OIL, AND GAS (WOG), NPTF-1 THREADS

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

The complete requirements for procuring the cock, shut-off, screw stem, threaded type, 150 psi, brass, water, oil, and gas (WOG), NPTF-1 threads described herein shall consist of this document and the issue in effect of CID A-A-59440.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This CID uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59440	/7	-	030
CID number	Applicable CID sheet (two digits)		Dash number (see figure 1)

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Cock, shut-off, screw stem, threaded type, 150 psi, brass, water, oil, and gas (WOG), NPTF-1 threads supplied to this CID shall be as specified herein (see figure 1).

General. This slash sheet covers the requirements for cock, shut-off, screw stem, threaded type, in nominal sizes of 1/8 inch through1/4 inch with NPTF-1 threads.

Intended application. This item is for use in liquid and gas systems where pressures do not exceed 150 psig (10.3 bar) water, oil and gas (WOG) at 150°F (65.6°C) and the liquid is compatible with brass.

Permissible variations in dimensions. Figure 1 lists the permissible variations in dimensions for each size. The external dimensions of the cocks should not limit procurement unless the end item applications restrict these dimensions.

Optional designs. The illustration is for identification purposes only is not intended to restrict designs or shapes not dimensioned.

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Materials.

Brass. Brass shall be in accordance with the following commercial standards:

ASTM B16	ASTM B121	ASTM B505	SAE AMS 4842
ASTM B22	ASTM B124	ASTM B584	SAE AMS 4845
ASTM B30	ASTM B148	ASTM B763	SAE AMS 4855
ASTM B36	ASTM B176	ASTM B770	SAE AMS 4860
ASTM B61	ASTM B271		SAE AMS 4862
ASTM B62	ASTM B369		SAE AMS 4890

Threads. Threads are dryseal NPTF-1 in accordance with CID A-A-59440.

NPTF-1 dryseal threads. NPTF-1 dryseal threads seal pressure tight joints without sealant compound. The engineering principle and design are explained in ANSI B1.20.3.

Handle. Handle may be steel hot rolled or cold rolled in accordance with QQ-S-698 or a commonly accepted industry standard.

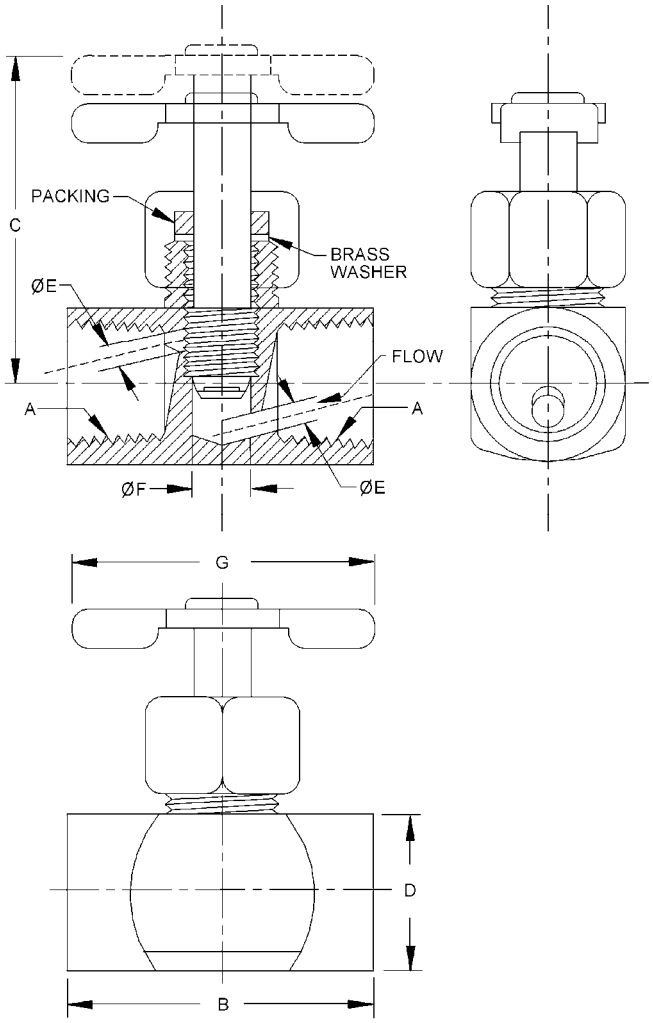
Protective coating.

a. Brass: None
b. Steel:
    (1) Cadmium in accordance with SAE AMS-QQ-P-416, type II, class 2.
    (2) Zinc in accordance with ASTM B633, type II, 12.7 µm (50 microinches) minimum.

Test pressures.

a. Shell 150 psig (10.3 bar).
b. Seat 50 psig (3.45 bar).

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Dash number	Handle plating	A pipe thread NPTF-1	B max (mm)	C max (mm)	D dia min (mm)	E dia min (mm)	F dia max (mm)	G max (mm)
030	Zinc	1/8-27	1.13 (28.7)	1.03 (26.1)	.50 (12.7)	.16 (4.0)	.19 (4.8)	1.39 (35.3)
031	Zinc	1/4-18	1.62 (41.1)	1.27 (32.3)	.69 (17.5)	.16 (4.0)	.19 (4.8)	1.39 (35.3)
032	Cadmium	1/8-27	1.13 (28.7)	1.03 (26.1)	.50 (12.7)	.16 (4.0)	.19 (4.8)	1.39 (35.3)
033	Cadmium	1/4-18	1.62 (41.1)	1.27 (32.3)	.69 (17.5)	.16 (4.0)	.19 (4.8)	1.39 (35.3)

FIGURE 1. Cock, plug, three way, threaded type.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .010 inch (0.25 mm).
4. Measurement across flats of hexagonal wrench pad may be a standard measurement.
5. Flow of liquid should be toward the valve seat not toward the stem threads.

FIGURE 1. Cock, plug, three way, threaded type - Continued.

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to CID. See classification information for PIN format example.

Source of documents.

Commercial Item Description

A-A-59440	-	Cocks, Drain, Cocks, Plug, and Cocks, Shutoff, Screw Stem; Brass, NPT and NPTF-1 Threads, General Requirements For

Federal Specification

QQ-S-698	-	Steel, Sheet and Strip, Low Carbon

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Other Publications

ASTM INTERNATIONAL

ASTM B16	-	Rod, Brass, Free-Cutting, Bar and Shapes for use in Screw Machines
ASTM B22	-	Castings, Bronze, for Bridges and Turntables
ASTM B30	-	Copper Alloys in Ingot Form
ASTM B36	-	Plate, Brass, Sheet, Strip, and Rolled Bar
ASTM B61	-	Castings, Steam or Valve Bronze
ASTM B62	-	Metal Castings, Composition Bronze or Ounce
ASTM B121	-	Plate, Leaded Brass, Sheet, Strip, and Rolled Bar
ASTM B124	-	Copper and Copper Alloy Forging Rod, Bar, and Shapes
ASTM B148	-	Aluminum-Bronze Sand Castings
ASTM B176	-	Die Castings, Copper-Alloy
ASTM B271	-	Copper-Base Alloy Centrifugal Castings
ASTM B369	-	Copper-Nickel Alloy Castings
ASTM B505	-	Castings, Copper Alloy, Continuous
ASTM B584	-	Castings, Copper Alloy Sand, for General Applications
ASTM B633	-	Zinc on Iron and Steel, Electrodeposited Coatings of
ASTM B763	-	Copper Alloy Sand Castings for Valve Application
ASTM B770	-	Copper-Beryllium Alloy Sand Castings for General Applications

(Copies of these documents are available from http://www.astm.org or ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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SAE INTERNATIONAL

SAE-AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)
SAE AMS4842	-	Castings, Leaded Bronze, Sand and Centrifugal 80CU - 10SN - 9.5PB as Cast
SAE AMS4845	-	Tin Bronze Castings, Sand and Centrifugal 87.5CU - 10SN - 2ZN as Cast
SAE AMS4855	-	Leaded Red Brass, Sand and Centrifugal Castings 85CU - 5.0SN - 5.0PB - 5.0ZN as Cast
SAE AMS4860	-	Manganese Bronze, Sand and Centrifugal Castings 58CU - 39ZN - 1.2FE - 1.0AL - 0.80MN as Cast
SAE AMS4862	-	Manganese Bronze, Sand and Centrifugal Castings 63CU - 24ZN - 6.2AL - 3.8MN - 3.0FE High Strength, as Cast
SAE AMS4890	-	Copper-Beryllium Alloy Castings 97CU - 2.1BE - 0.52CO - 0.28SI Solution Heat Treated (TBOO)

(Copies of these documents are available from http://www.sae.org/ or SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
79470	Eaton Corp 14615 Loneoak Rd. Edenperry, MN 55344 952-294-2032/888-258-0222 jeffenoe@eaton.com
93061	Parker-Hannifin Corp Brass Products Division 300 Parker Drive Otsego, MI 49708-1472 260-636-2104 dwsanders@Parker.com
20969	Essex Brass Corp 23500 Pinewood Warren, MI 48091-3122 586-757-8200 bpushman@essexbrass.com

Military Standard (MS) replacement data. This CID replaces the following Military Standard as shown in table I.

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TABLE I. MS supersession data.

CID PIN	Superseded MS
AA59440/7-030 Steel (zinc plated)	MS35934-1
AA59440/7-031 Steel (zinc plated)	MS35934-2
AA59440/7-032 Steel (Cadmium plated)	MS35934-1
AA59440/7-033 Steel (Cadmium plated)	MS35934-2

1/ These parts were originally listed in MS35935 as a single part type. Due to special handling/disposal and corrosion resistance requirements the parts have been separated in this CID.

Part number (P/N) supersession data. These CID P/N's supersede the following MFR's P/Ns as shown in table II. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. PIN supersession data.

CID PIN	MFR's CAGE	MFG's P/N 1/
AA559440/7-030	01276	6800
AA559440/7-031	01276	6805
AA559440/7-032		
AA559440/7-033		

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID. At the time of preparation of this CID, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements see the marking paragraph.

Marginal notations. Marginal notations are not used in this revision to identify changes due to the extent of the changes.

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - SH
Air Force - 99
DLA - CC

Review activities:
Army - CR4, GL, MI
Navy - MC

CIVIL AGENCY AND COORDINATING ACTIVITIES:

GSA - FSS

Preparing activity:

DLA - CC

Project 4820-0885-000

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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